

Balloumi Karim

Software Engineer

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Professional Profile

Software Engineer with 3+ years of experience building scalable web and mobile applications. Strong background in full-stack development, clean architecture, and AI-driven solutions, including RAG systems and local LLM integration.

Motivated to contribute to robust, secure, and maintainable software systems within an innovative European engineering team.

Primary stack: JavaScript, PHP, Django, React.js, MySQL, REST APIs, RAG, Local LLMs, LangChain, LangGraph, n8n

Work History

01/2026 – 03/2026 **Software engineer for Mobile and web Application**

GreenGuardian: AI-Powered Mobile Ecokids Educational platform for environmental protection and arboriculture.

Tech stack: Flutter, Firebase, Qwen 8B (Local LLM), RAG

- Integrated a local Qwen 8B model to enable offline AI functionality, ensuring 100% data privacy and zero API latency for users.
- Developed a RAG-based expert system to provide scientifically accurate advice on tree care by processing specialized botanical datasets.
- Engineered a Computer Vision module that allows users to scan trees for real-time health diagnosis and hydration alerts.
- Implemented AI Voice Chat and Gamification features, increasing user engagement among younger demographics by providing interactive, accessible learning tools.

01/2024 – 06/2024 FlutterFlow Developer Intern RIATICA

- Developed a comprehensive mobile management application using FlutterFlow, reducing development time by 40% compared to traditional coding.
- Integrated Firebase for real-time database management and secure user authentication.
- Designed and implemented high-fidelity UI/UX layouts and conducted rigorous user testing to ensure seamless performance.

06/2023 – 08/2023 Full Stack Web Developer Intern Cloudini

- Built a full-scale e-commerce platform using React.js, PHP, and MySQL.
- Developed core features including dynamic product catalogs, shopping cart logic, and secure online payment gateways.
- Designed a custom administrative dashboard for inventory and order management.

01/2022 – 04/2022 IT Maintenance Intern Smart Tunisian Technoparks

- Designed and implemented a preventative maintenance calendar to ensure 99% hardware uptime.
- Prioritized technical interventions and collaborated with multidisciplinary teams to resolve network issues.

Education

2024 – Present | Engineering Degree in Software Engineering SESAME University (Private School of Engineering)

- Focus: Advanced Software Architecture, System Design, and Algorithmic Optimization.

2021 – 2024 | Bachelor's Degree (Licence) in Web Development & Multimedia Higher Institute of Technological Studies (ISET)

- Core Studies: Full-stack web development, multimedia integration, and database management.

Languages

- French: Native (B1)
- English: Professional Working Proficiency (B2)